NEEL GHOSHAL, M.S.

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EDUCATION

M.S. Computer Science | NC State University, Raleigh, NC | Aug 2024 - Aug 2026

- Thesis: Investigating the Impact of Technical Debt Remediation on Code Translation Tasks: four-stage pipeline across frontier models including GPT-4/ChatGPT, Claude, Gemini, and DeepSeek; 70 controlled experiments; custom statistical evaluation framework
- Coursework: Optimization and Algorithms, Advanced Deep Learning, AI, Statistics, Independent Study: ML in Software Engineering

B.E. Computer Science Engineering | Vellore Institute of Technology, India | Jul 2020 - May 2024

- Capstone: Indic Legal LLM: production LLM system with NLP

TECHNICAL SKILLS

Data Science and Statistics: Python (expert), SQL (proficient), R (familiar); statistical methods, hypothesis testing, predictive modeling, exploratory data analysis, feature selection and engineering, model validation and optimization; data quality and governance; Matplotlib, Seaborn, Plotly for visualization and reporting
Machine Learning and AI: scikit-learn, PyTorch, TensorFlow, XGBoost, FAISS, RAG, LangGraph, MCP, multi-agent systems, NLP, Generative AI, LLMs (OpenAI, Anthropic, Gemini APIs); Weights and Biases; Pandas, NumPy, scipy, statsmodels; model evaluation and benchmarking
Engineering and Collaboration: Golang, Java, C++ (proficient); TypeScript/JavaScript, FastAPI, AWS (familiar), Snowflake Cortex; Django, REST APIs; Git, Docker, Linux; cross-functional collaboration translating requirements into analytical solutions; mentoring and knowledge-sharing

WORK EXPERIENCE

Machine Learning Engineer (Intern) | GeoForAll Lab, NC State University | Jun 2025 - Jul 2025

Raleigh, NC

Tech: Python, PyTorch, FAISS, Qwen2.5-Coder-7B, RAG

- Applied statistical and machine learning techniques to develop a natural language interface over 200+ real-world datasets: built a RAG pipeline on Qwen2.5-Coder-7B with FAISS-based semantic retrieval, token-budgeted context, and constrained two-step generation with output validation
- Designed an evaluation framework with exact-match and partial-credit scoring, triaging cases where the metric disagreed with expert judgment to separate real improvements from noise; achieved 5x performance improvement and 30% reduction in computational overhead; communicated findings to non-technical stakeholders

Research Assistant | Innovative Education Computing Lab, NC State University | Sep 2024 - Dec 2024

Raleigh, NC

Tech: Python, Pix2Pix GAN, MMdetection

- Built a data generation pipeline converting simulation imagery into realistic, labeled training data: used a Pix2Pix GAN for image-to-image translation and MMdetection for bubble segmentation and masking, producing a labeled dataset for a downstream neural network evaluating nuclear power plant operations from bubble imagery; collaborated with senior researchers to improve analytical workflow efficiency by 80%

Data Science Intern | Prayojaka Technologies | Apr 2024 - May 2024

Tech: Python, Pandas

- Reviewed and analyzed prior data science work in the context of current company projects, identified trends and data quality issues, and communicated actionable findings to stakeholders

Machine Learning Intern | Reliance Jio | May 2023 - Jul 2023

Tech: Python, Big Data, SQL

- Built and validated predictive models over a 200K+ user dataset; performed exploratory data analysis, feature engineering, and statistical analysis on large-scale structured data; ensured data quality throughout the pipeline; reported findings to cross-functional stakeholders

Blockchain Developer Intern | Evia Network | Aug 2022 - Oct 2022

Tech: Golang, Cosmos SDK, REST APIs

- Designed and implemented distributed systems using Cosmos SDK and Golang, built REST API interfaces, and collaborated cross-functionally on system architecture and feature development

PROJECT EXPERIENCE

Veritas: Multi-Agent Document Integrity Review | MLH Best Use of Snowflake Winner | 2026

github.com/NeelGhoshal/Eval_Agent_System_NRTF

Tech: Python, LangGraph, Snowflake Cortex, Streamlit, Sentry

- Co-built a LangGraph system in which four parallel agents review a bill or article for completeness, factual claims, actor authority, and internal contradictions; LLM calls run through Snowflake Cortex, and claim and authority verdicts cite web-sourced evidence
- Each agent writes only to its own state key so one failure does not block the others; Streamlit report UI with Sentry error monitoring that excludes pasted content

ToolHelper: Token-Efficient MCP Tool Calls | 2026

github.com/NeelGhoshal/ToolHelper

Tech: Python, MCP, LLM Agents, Ollama, Streamlit, Experiment Design

- Built a monitoring layer for LLM agent tool calls on three biomedical MCP servers (Medical MCP, BioMCP, OMOP MCP): exact tokens per call and the pathway to each tool
- Tested schema compression, result shaping, a turn limit with forced final answer, and an "answer early" instruction in fractional factorial experiments on a 54-task benchmark with a versioned decision log; reduced tokens per successful task by roughly 50% (about 52,300 to 25,000) while success rose from 88% to 96%

Cedarline Insurance Service Agent: Agentforce | Oct 2026

github.com/NeelGhoshal/Insurance_Service_Agent

Tech: Python, Pandas, Salesforce Data Cloud, Agentforce Builder

- Built a member service agent for a hypothetical health plan that answers questions on claims, denials, deductibles, prior authorizations, and appeals; routes across subagents for identity verification, member services, plan information, and escalation
- Built the data pipeline: mapped Synthea patients, payers, and claims into a synthetic dataset (1,100+ members, 5,500+ claims) and loaded it into Snowflake and Salesforce Data Cloud; added guardrails for member scoping, no medical advice, and prompt-injection test cases

Paper Librarian: MCP Server for Research Papers | Oct 2026

github.com/NeelGhoshal/MCP

Tech: Python, MCP Python SDK, SQLite (sqlite-vec, FTS5), sentence-transformers, PyMuPDF, pytest

- Built an MCP server giving Claude a local research-paper library: 8 tools (hybrid search, paper comparison, citation graph, notes), PDF and arXiv ingestion, and hybrid vector and BM25 retrieval with rank fusion, with answers citing paper, section, and page; 90 pytest tests

PUBLICATIONS, CERTIFICATIONS AND LEADERSHIP

Publications (6): Chest X-Ray Classification (2023); AI in Radiation Oncology (2023); Mental Health Chatbot with NLP and Deep Learning (2023); AI in Solar Systems (2024); Mental Health Chatbot with Data Augmentation (2024); Real-Time Visual Data Processing using Neuromorphic Systems (2025)
Certifications: IBM (Coursera): Fundamentals of Building AI Agents, Agentic AI with LangChain and LangGraph | Statistics | ML Stanford/DeepLearning.AI | AI Analyst | TF Developer | Salesforce AgentBlazer Champion
Leadership: President, Ratio Christi NC State (2025-2026): hosted events with 100+ participants, Legatus Christi Award | Co-Lead, Cohere for AI (50+ participants): mentored workshop participants | VP, CodeChef-VIT (4,000+ participants)